

MULUND COLLEGE OF COMMERCE (AUTONOMOUS)

Programme: School of Business- Business Administration Class: TYBBA Semester: V

Exam: External

Name of the Subject: Optimization Techniques

Date of Exam: 06/10/2025

Marks: 60

Time: 2 Hours

Q1 Answer the following

i) A company uses annually 48000 units of raw material costing Rs. 1.20 per unit. The ordering cost is Rs. 45 and carrying cost are 15% per year of the average inventory. Find EOQ. **08**

A ii) Find the best time to replace the machinery costing Rs. 15000. The maintenance cost and resale value are as given below

Year	1	2	3	4	5	6	7	8
Running Cost	1000	5000	2000	2500	3000	4000	4500	5000
Resale value	13000	12000	11500	10500	6000	7500	6000	5000

07

OR

iii) The initial cost of a machine is 3000, running cost is as given below

Year	1	2	3	4	5	6	7
Running cost	500	600	800	1000	1300	1600	2000

08

B It is given that the rate of interest is 10% & scrap value is nil. When should the machine be replaced?

iv) The production company requires 7200 kg of material for manufacturing per year. The cost of ordering is Rs.72 and the cost of carrying inventory is 25% . The price is Rs. 20 per kg. Find the beat ordering quantity. Also find Total cost **07**

Q2 Answer the following

i) Solve the following LPP using simplex method.
Maximize $Z = 31x_1 + 42x_2 + 13x_3$. Subject to
 $x_1 + x_2 + 2x_3 \leq 40$; $2x_1 + x_2 + 5x_3 \leq 50$; $2x_1 + x_2 + x_3 \leq 35$; $x_1, x_2, x_3 \geq 0$ **08**

A ii) The simplex table for maximization problem is given below

			15	30	40	0	0	0
Cj	BV	BS	X1	X2	X3	S1	S2	S3
40	X3	8	0	1	1	1	0	0
15	X1	16	1	-1	0	-1	1	0
0	S3	4	0	0	0	-1	-1	1

07

Is the above solution optimal? If not find the optimal solution.

OR

B Solve the LPP using Simplex Method
Minimize $Z = 40x_1 + 24x_2 + 50x_3$ subject to **15**
 $x_1 + 2x_3 \geq 20$, $2x_1 + 1x_2 \leq 25$, $4x_1 + 2x_3 \geq 15$, $x_1 \geq 0$, $x_2 \geq 0$, $x_3 \geq 0$

Q3**Answer the following**

- i) Solve the following Assignment Problem to minimize the cost

	M1	M2	M3	M4
P1	20	90	20	70
P2	60	80	70	60
P3	40	60	50	30
P4	40	20	70	30
P5	50	30	90	50

08**A**

- ii) Find IFS by a) North-West corner method and b) Least Cost Method

	M1	M2	M3	M4	Supply
W1	16	13	15	14	2000
W2	15	19	12	17	1500
W3	15	17	18	16	1000
Requirement	700	1200	1700	900	

07**OR**

- iii) Solve the following Assignment Problem to maximize the Profit

	S1	S2	S3	S4
T1	16	11	13	14
T2	17	10	15	16
T3	14	10	15	13
T4	13	11	12	10
T5	14	15	13	11

08**B**

- i)

- iv) The following table shows transportation cost and solution as well.

	D1	D2	D3	D4	Supply
F1	5	3 (50)	2	7	50
F2	7	8	1 (45)	5 (15)	60
F3	3 (70)	5 (5)	7	9 (15)	90
Demand	70	55	45	30	

07

- a) Is this solution feasible?
 b) Is this solution optimum, if not find the optimum solution?

Q4**Answer the following**

i) For the following is the data about a project

Activity	A	B	C	D	E	F	G
Predecessors	--	--	A	A,B	C,D	B,D	E,F
Duration (days)	7	6	8	7	6	8	6

08

Draw network diagram, find critical path, total float and free float for each activity

A

ii) A project has the following activities and other characteristics

	Activity	1 – 2	1 – 3	1 – 4	2 – 5	2 – 6	3 – 6	4 – 7	5 – 7	6 – 7
Time	Optimistic	5	18	26	16	15	6	7	7	3
	Pessimistic	13	22	40	20	25	12	13	9	5
	Most likely	6	20	33	18	20	9	10	8	4

07

Draw network diagram. Find expected length and standard deviation of critical path.

OR

The following table shows the activities in a project and other relevant information.

	Activity	A	B	C	D	E	F	G
	Predecessors	--	--	--	A	C	A	B, D, E
Time (days)	Normal	4	6	2	5	2	7	4
	Crash	3	4	1	3	2	5	2
Cost (Rs.)	Normal	60	150	38	150	100	115	100
	Crash	90	250	60	250	100	175	240

B**15**

Draw network diagram, determine the total cost to complete the project in 10 days.